

Units matter: cooperation in large language model agents tracks how the payoff matrix is written, not what it means

Author One¹ and Author Two^{1,2}

¹Affiliation One

²Affiliation Two

Abstract

Multiplying every payoff by a positive constant leaves the outcome ordering, the dominant action, the equilibrium set and the replicator dynamics untouched, so for any agent playing the game the predicted effect is zero by theorem rather than by convention. We apply it to seven language models in the iterated prisoner’s dilemma, across six payoff scales spanning five orders of magnitude, five prompt languages and 12,000 agent-games. Cooperation moves in six of the seven, by up to 0.32, and not monotonically: lowest where the cells print as decimals, highest where they print as short integers, intermediate where they print as thousands. A three-level factor built from how the numbers are written beats any polynomial in the log scale and matches one saturated in it. The effect is already open on the first move, before any history exists. It reaches the strategy label selectively, moving the unconditional rules but not the reciprocal ones or the coordination structure, and shifts how far a persona instruction is obeyed in every model. Two frontier models change places in the cooperation ranking, so a cooperation rate reported for a language model is a statement about one arbitrary choice of units, and should be published with its scale sensitivity.

Keywords: machine behaviour; large language models; iterated prisoner’s dilemma; payoff scaling; utility invariance; benchmark validity

1 Introduction

Language models are now deployed as *agents* that negotiate, coordinate and decide on behalf of their users [1, 2], and a fast-growing literature measures how they behave when they do. The instrument of choice is the classical game, and the prisoner’s dilemma above all, because its strategy space is mapped and its predictions are sharp [3–7]. Studies in this vein report cooperation rates, departures from Nash play, model-specific signatures and archetypes recovered from long games [8–12], and the numbers they produce increasingly feed models of mixed human-machine populations [13–15].

Every one of those measurements is made at a particular payoff matrix. The matrix is written down once, in whatever units the experimenter chose, and the cooperation rate is then reported as a property of the model. The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, and

such a utility is defined only up to a positive affine transformation [16, 17]. Writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no replicator trajectory. Under the theory the two matrices are not two games written differently. They are one game.

That makes a positive rescaling the sharpest possible probe of whether a language model is playing the game it was shown. Most manipulations one might apply to a game-theoretic prompt have arguable effects: a change of framing carries new words, a change of persona carries new information about the agent, a change of language carries a different training distribution. A rescaling carries none of these. It changes the numerals and nothing else. If behaviour is a function of the game, the effect is exactly zero, and the null needs no defending because it is a theorem. If behaviour moves, then whatever the model is responding to is not the game.

This paper takes the rescaling itself as its object of study. We apply it across five orders of magnitude to seven models in the iterated prisoner’s dilemma, crossed with five prompt languages and two persona instructions, and ask four questions.

1. Does the strategic behaviour of a language model survive multiplying every payoff by a positive constant, and by how much does it fail to?
2. If it moves, what does it move with? A rescaling changes the magnitude of the numbers and the way they are printed at the same time. These are different functions of λ , and the data can tell them apart.
3. Where in the game does the effect enter: at the opening move, before any history exists, or through the course of play?
4. What else does the scale carry with it, and what does it leave alone?

Our contributions are the following. **(i)** We report a response curve over six payoff scales from 10^{-2} to 10^3 , and it is not monotone: it rises, plateaus and falls (§3.1). **(ii)** We show that the response is better described by a three-level factor built from how the cells are *printed* than by any polynomial in $\log_{10} \lambda$, and no worse than a factor saturated in λ itself, at three fewer degrees of freedom (§3.2). **(iii)** We show that the effect is already open on the opening move, and that what moves is the level of cooperation and not the coordination structure of play, whose miscoordination rate is flat across all six scales (§3.4–§3.5). **(iv)** We show that the payoff scale shifts how far a persona instruction is obeyed, in the same direction in every one of the seven models, while leaving the sign of that effect a property of the model; and that the scale interacts with the prompt language in both arms (§3.6–§3.7). **(v)** We show that the strategy a read-out assigns to a run of play moves with the scale in all seven models, and that it moves on the unconditional rules while the conditional ones stay nearly still, which is the dissociation the coordination measure records read at the level of the sequence (§3.8). **(vi)** We show that the choice of λ reverses the cooperation ranking of two frontier models, and propose a reporting standard that costs one extra condition (§3.3, §4.7).

An earlier preprint by the present authors reported that the cooperation rate of open-weight models varies with the payoff scale and with the prompt language [18]. That report established the phenomenon in one arm of the corpus. The present paper asks what the phenomenon is a function of, and every contribution numbered above is new to it: the frontier replication, the magnitude-against-notation comparison, the localisation of the effect at the opening move, the

persona result, the coordination invariant, the strategy read-out, the ranking reversals and the reporting standard. The corpus is the same, re-analysed under the payoff polarity fixed in §2.2.

2 Material and methods

2.1 The invariance under test

Let u be a von Neumann–Morgenstern utility over the four outcomes of a two-by-two game. The representation theorem says that u and $a + bu$ with $b > 0$ describe the same preferences [16, 17]. Every quantity a game theorist computes from the matrix is therefore invariant under that map: the ordering of the outcomes, the best-reply correspondence, the set of Nash equilibria, the dilemma indices greed $(T - R)/(T - S)$ and fear $(P - S)/(T - S)$, and the replicator flow, which is invariant under b up to a rescaling of time and under a exactly.

We manipulate the multiplicative half of that map, $u \mapsto \lambda u$ with $\lambda > 0$, and we say so explicitly because the two halves are not equally demanding. The additive half moves every cell by the same amount and can be absorbed into a change of reference point; the multiplicative half changes the number of digits in every cell and is therefore the half that a system reading the matrix as text is most likely to notice. We do not test the additive half here, and we note in §4.6 what that leaves open.

Figure 1a states the stage game and the list of quantities that λ cannot touch. Figure 1b states what it does touch, which is the string the prompt prints.

2.2 The game, the two arms and the crossing

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [19] provides the computational infrastructure. Conditions are written in JSON; at run time the framework combines them with language-specific prompt templates, simulates repeated normal-form games and logs every round of play. We adopt the framework and its multilingual templates as given, which keeps the wording of the manipulations out of our hands, and we add the payoff scaling.

Agents played a symmetric two-player prisoner’s dilemma. The templates state the cell values as *penalties*, that is, years of imprisonment in the prisoner’s-dilemma story, and instruct the agent to *minimise* them, so a smaller number is a better outcome and the ordering is $T < R < P < S$. That is the penalty-space image of the usual $T > R > P > S$, and it makes Option A the dominant, defecting action and Option B cooperation. We state the direction explicitly because reversing it would reverse the sign of every result below. The assignment is not a matter of interpretation: FAIRGAME logs the realised payoff next to every action, and the assignment above reproduces the logged payoff on all 264,000 rounds of the corpus while the opposite assignment reproduces none of them.

The corpus has two arms (figure 1c). The *open-weight arm* is the one that carries the scale sweep. Three models, Gemma 3 12B, Llama 3.1 8B and Qwen3 8B, each met six payoff scales, $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$, five prompt languages and four ordered persona pairings, over thirty rounds. The *frontier arm* replicates the design at a shorter horizon and a narrower sweep. Four models, Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large, met three scales, $\lambda \in \{0.1, 1, 10\}$, the same five languages and the same four pairings, over ten rounds. Both arms are balanced at exactly 80 agent-games in every model $\times$ language $\times$ scale cell, with none missing. That gives 3,600 dyads and 7,200 agent-games in the open-weight arm and

2,400 dyads and 4,800 agent-games in the frontier arm, for 6,000 dyads, 12,000 agent-games and 264,000 binary decisions in all.

The base matrices differ between the arms. The frontier arm used $T = 0$, $R = 2$, $P = 6$, $S = 10$, which is greed 0.20 and fear 0.40 on the utility scale $u = -\text{penalty}$; the open-weight arm used $P = 8$, which is greed 0.20 and fear 0.20. The two arms are therefore two dilemmas rather than one, and we never pool them. Each is internally a single game shown at several scales, which is the comparison every result below is made within.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players’ choices, and returned a single token naming one of two options. Communication was disabled, so the only channel between agents is the action history. Actions carry the labels `OptionA` and `OptionB` in all five languages, which avoids the extra meaning that “cooperate” and “defect” would bring; the templates never say which label means cooperation, and the agent has to read it off the payoff table. That is what makes the payoff numbers the load-bearing part of the prompt, and it is why a change to them is the manipulation of interest.

Decoding used a fixed low output-token budget with reasoning traces disabled. That setting is not a detail. A model that emits hidden reasoning can exhaust a small budget inside the trace and return an incomplete string, and a harness that then falls back to a default action turns that string into an apparent cooperation rate of 1.0 that the parser has created rather than the model. We logged parse-failure and fallback rates for every cell and required both to stay below 0.02. Every dyad ran its full horizon and every decision resolved to one of the two option tokens, so no cell loses games to a failure to parse.

2.3 Notation regimes

Because λ changes the magnitude of the payoffs and the way they are printed at the same time, we define the printed quantities in advance and let the model comparison of §3.2 arbitrate between them. For each scale we take the four cell values as the template renders them and record three features: whether any of them carries a decimal point, the mean number of glyphs across the four, and a three-level *notation regime* that partitions the sweep into *fractional* ($\lambda < 1$, at least one cell printed with a decimal point), *unit* ($\lambda \in \{1, 10\}$, largest cell two or three digits) and *large* ($\lambda \geq 100$, largest cell four or five digits). Figure 1b lists the six rows.

One property of that partition has to be stated before any of it is interpreted. In this design $\lambda < 1$ and “the cell prints with a decimal point” are the same event, so a fractional-notation account and an account based on sub-unit magnitude make identical predictions here and cannot be separated. What the design *can* separate is either of them from a smooth function of $\log_{10} \lambda$, and that is the comparison we run. We return to the experiment that would break the remaining confound in §4.6.

2.4 The strategy read-out, and where its labels come from

A cooperation rate averages over a game. The second summary the literature reports, and the one that feeds evolutionary models, is the strategy assigned to a whole run of play, so we measure that too. Because the assignment is made by an instrument rather than observed, the instrument is described here in full and its output is reported by provenance rather than as a single label column.

The vocabulary is the four canonical memory-one rules: AllC, AllD, tit-for-tat [3] and

win-stay lose-shift [20]. The read-out runs in two stages. Every one of the four prescribes an action as a function of the previous round’s joint outcome alone, so the first stage checks by enumeration whether a trajectory is *exactly* what a rule would have played, and returns the set of rules that fit; no learning is involved and the answer is a proof. The first stage has three outcomes: exactly one rule fits, and the label is provable; several fit, in which case the run is recorded as *ambiguous*, which happens whenever the realised history never discriminates them, as when an agent cooperates throughout against a cooperator and is thereby AllC, TFT and WSLS at once; or none fits. Where none fits, a second stage supplies the nearest rule: a single-layer long short-term memory network [21] with a 24-dimensional embedding and 96 hidden units, trained on synthetic play generated from the same four rules at zero and five per cent execution noise, returns a posterior over them and we take its argmax. Separate networks are trained for the two horizons, and each arm is read with the network matching its own.

Two properties of that pipeline bear on how its output should be read, and both are reported rather than assumed away. The first is provenance. Over the open-weight arm 22.1% of agent-games carry a provable rule, 11.8% are ambiguous and 66.1% are named by the network; over the frontier arm the shares are 25.5%, 8.3% and 66.2%. Two thirds of the labels are therefore inferences and not deductions, and of those a further 16.5% and 25.7% of all agent-games sit below a posterior of 0.90, a floor at which a read-out designed to abstain would decline to answer at all. The second is validation. On held-out synthetic play the pipeline recovers the generating rule with accuracy 0.975 at ten rounds and 0.953 at thirty, against Bayes-optimal single-label ceilings of 0.987 and 0.966, and where a rule fires uniquely the label is right 99.5% and 98.7% of the time. The open-weight arm sits almost exactly at that validated operating point; the frontier arm departs from the canonical rules by more than the training distribution does, so its labels are nearest neighbours more often. The electronic supplementary material, §S1, gives the full validation, including how the network degrades on noise levels it never saw.

We therefore treat the strategy mix as a second, coarser dependent variable rather than as ground truth, and we report every test on it twice: once over the five categories the read-out emits, and once over the four named rules alone. The distinction matters because *ambiguous* is a statement about how identifiable the play is and not about which strategy was played, so a shift into or out of that category is a shift in nameability. Where the two versions of a test disagree we say so.

2.5 Statistical analysis

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [22]. The choice of unit matters: rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would narrow the interval by roughly a factor of $\sqrt{2}$.

Whether a factor moves a measure at all is answered against a permutation null that shuffles the factor label across *dyads*, so a dyad keeps both of its agent-rows together and the null preserves the dependence the design creates. We report the observed largest between-level gap, the permutation P value over 2,000 draws, and the null’s 95th percentile, which doubles as the smallest shift the test could have detected. For distributions over strategy labels we use the largest total-variation distance between two levels, with the same dyad-level null.

Whether the response is a function of the magnitude or of the notation is a model-comparison

question, and we answer it with the Bayesian information criterion [23] over a ladder of specifications fitted to identical rows. Every specification carries the same controls for model, language, own persona and opponent persona, and they differ only in how λ enters: not at all; linearly, quadratically or cubically in $\log_{10} \lambda$; as the fractional flag; as the fractional flag plus the glyph count; as the three-level notation regime; or saturated, as a free level per scale. The saturated specification is the upper bound on fit that any function of λ can reach, so a description that matches it at fewer parameters is a description that has found the right shape.

We do not correct for multiple comparisons across models, and we say so rather than leave the point to be inferred. Each model is a separate system and is its own family, so the tests within a manipulation are seven families of one rather than one family of seven. The claims we draw are in any case about whether the rescaling moves behaviour at all, not about which model moves most.

2.6 Confounds we declare

Three features of the design are confounded with the arm, and one with a single cell. The two arms differ in horizon (thirty rounds against ten), in whether the round count was disclosed (it was in the open-weight arm, and in the frontier arm only for Gemini), in sampling temperature, which follows each provider’s recommended default as in the FAIRGAME protocol, and in the base matrix as recorded in §2.2. All four are constant within a model, so none of them can generate a within-model effect of λ , which is what every result below is. They do mean that no coverage or level statistic is comparable across arms, and we compare none.

Separately, four dyads in the Claude $\times$ English $\times$ $\lambda = 10$ cell are logged with the payoffs of $\lambda = 1$. We recovered the realised scale of every game from the logged values rather than assuming it from the directory name, which is how we found them; they are 8 agent-games out of 12,000, and dropping them moves Claude’s cooperation rate at $\lambda = 10$ from 0.708 to 0.715. We retain them at their nominal scale and note that no conclusion turns on them.

3 Results

3.1 A rescaling that cannot matter, and does

Multiplying every payoff by a positive constant moves the cooperation rate (figure 2). Pooled over the open-weight arm it runs 0.357 [0.332, 0.382] at $\lambda = 0.01$, 0.384 [0.357, 0.410] at 0.1, 0.573 [0.547, 0.599] at 1, 0.567 [0.542, 0.592] at 10, 0.476 [0.451, 0.503] at 100 and 0.502 [0.477, 0.529] at 1000. Pooled over the frontier arm it runs 0.489 [0.467, 0.510], 0.633 [0.616, 0.651] and 0.631 [0.612, 0.649] across its three scales. The same game, written with a different number of digits, is played differently.

Per model the largest shift between two scales is 0.316 for Gemma 3 12B, 0.267 for Qwen3 8B, 0.198 for Llama 3.1 8B, 0.240 for Claude 3.5 Haiku, 0.234 for GPT-4o, 0.080 for Gemini 3.5 Flash-Lite and 0.036 for Mistral Large. All but the last exceed a permutation null that shuffles the scale label across dyads: $P < 0.001$ for the first five and $P = 0.0075$ for Gemini, against null 95th percentiles between 0.047 and 0.121 (table 3). Mistral Large is the exception, at $P = 0.38$, and we report it as one: on this measure its behaviour is invariant to the rescaling in the way the theory says it should be. Its strategy mix is not, and we return to that in §3.8.

The shape of the curve is the first substantive result, and it is only visible because the

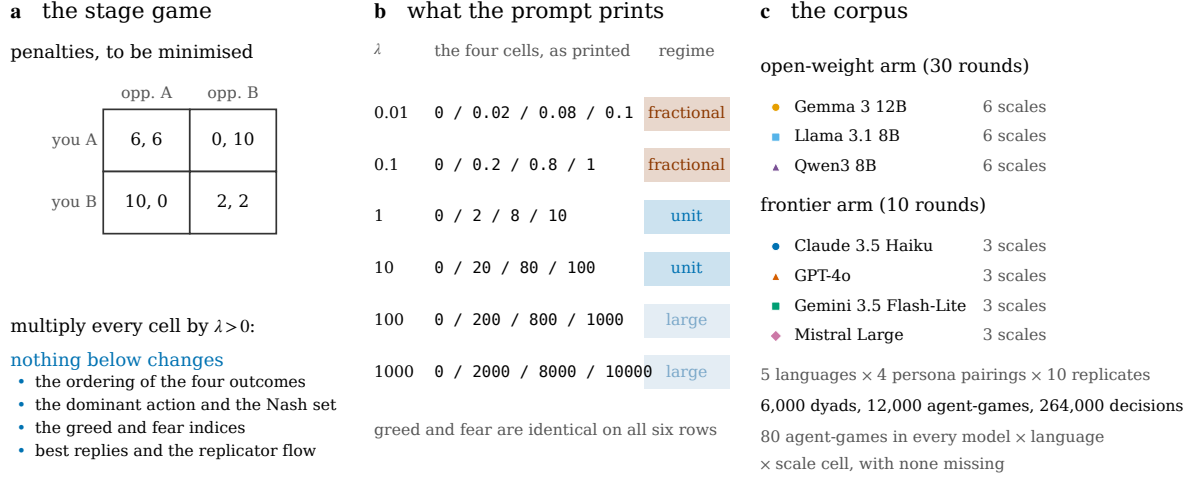


Figure 1: The game, the rescaling and the corpus. (a) The stage game of the frontier arm at $\lambda = 1$. Cells are penalties the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Multiplying every cell by $\lambda > 0$ is the multiplicative half of a positive affine transformation of a von Neumann–Morgenstern utility, and leaves every listed quantity where it was. The open-weight arm uses the same matrix with $P = 8$, which is fear 0.20 rather than 0.40. (b) What the rescaling does change: the four cell values as the prompt prints them, at each of the six scales of the open-weight arm, with the three-level notation regime of §2.3 on the right. Greed and fear are identical on all six rows. (c) The corpus. Both arms are balanced at 80 agent-games in every model $\times$ language $\times$ scale cell.

open-weight arm sweeps five orders of magnitude. The response is not monotone. In all three open-weight models cooperation rises from its minimum in the fractional regime to a maximum at $\lambda = 1$ or $\lambda = 10$ and then falls again by $\lambda = 100$: Gemma from 0.345 to 0.661 and back to 0.512, Qwen3 from 0.334 to 0.601 and back to 0.465, Llama from 0.300 to 0.499 and back to 0.450. A three-scale window at $\{0.1, 1, 10\}$, which is what the frontier arm and most published designs sample, lies entirely on the rising limb and would support the conclusion that larger payoffs produce more cooperation. Over the wider sweep that conclusion is false.

3.2 Magnitude or notation

The response tracks the notation. Pooled over the open-weight arm, cooperation in the fractional regime is 0.370 [0.352, 0.388], in the unit regime 0.570 [0.553, 0.588] and in the large regime 0.489 [0.472, 0.508] (figure 3a); the same ordering holds separately in all three models. In the frontier arm, which has no large regime, cooperation is 0.489 [0.467, 0.510] in the fractional regime against 0.632 [0.619, 0.644] in the unit regime, and the ordering again holds separately in every model.

The description ladder makes the point quantitatively (figure 3b, table 4). On the open-weight arm, against the best specification, a model with no λ term at all pays 374 BIC points, a linear term in $\log_{10} \lambda$ pays 247, a quadratic pays 110 and a cubic pays 118. A single binary feature, whether the cell prints with a decimal point, pays 56, and adding the glyph count brings that to 16. The best specification is the three-level notation regime, and the specification saturated in λ , which spends three more parameters and is the ceiling on what any function of λ can fit, pays 20 points more than it. Two features of how the numbers are written therefore capture what a free parameter per scale captures, while a smooth function of the magnitude, at any of the three

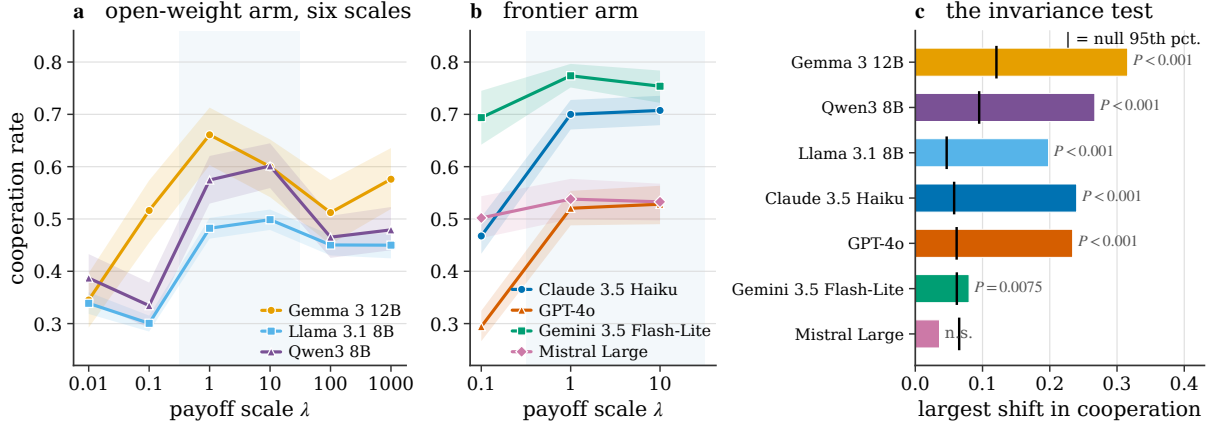


Figure 2: Cooperation against the payoff scale. (a) The open-weight arm over six scales. Bands are 95% bootstrap intervals over 2,000 resamples of whole dyads. The shaded strip marks the unit regime. The curve is not monotone in any of the three models: it rises to a maximum at $\lambda = 1$ or $\lambda = 10$ and falls again by $\lambda = 100$. (b) The frontier arm over three scales, which sample only the rising limb. (c) The invariance test. Bars give the largest shift in cooperation between two scales; ticks give the 95th percentile of a permutation null that shuffles the scale label across whole dyads. The shift a theory of the game predicts is exactly zero. Six of the seven models exceed their null.

orders we fitted, does not.

The frontier arm agrees where it can. With only two regimes present, the fractional flag and the notation regime are the same one-parameter specification there, and it beats the quadratic and the saturated model alike by 8.4 BIC points, and the linear term by 53.7.

Per model (electronic supplementary material, table S3), five of the seven prefer a notation description: Llama and Qwen3 in the open-weight arm, by 258 and 149 BIC points over the quadratic, and Claude, GPT-4o and Gemini in the frontier arm. The two exceptions run in opposite directions and we report both, including the one that runs against us. Gemma 3 12B prefers a cubic in $\log_{10} \lambda$, with the saturated specification 3.3 points behind it, the fractional flag plus glyph count 9.2 behind and the notation regime 24.3 behind; for that model a smooth function of the magnitude is the better description, and it is the single case in the corpus where the notation account loses to a polynomial. Mistral Large prefers no λ term at all, which is the correct description of a model whose cooperation rate did not move.

3.3 Which model looks more cooperative depends on the units

The consequence for measurement is direct (figure 3c). Within the open-weight arm, the smallest scale swing inside a model, 0.198 for Llama, is larger than the median spread between the three models at a fixed scale, 0.114. Which λ the experimenter wrote down therefore matters more, for the number that comes out, than which of the three models was asked. The frontier arm does not reproduce that ordering, because its models are further apart: there the median between-model spread is 0.253 against a largest within-model swing of 0.240.

Rankings reverse. Of the 45 model-pair by scale-pair comparisons available in the open-weight arm, 8 change sign, and of the 18 available in the frontier arm, 2 do. Concretely, Qwen3 8B cooperates more than Gemma 3 12B at $\lambda = 0.01$ (0.387 against 0.345) and less at every larger scale, reaching 0.574 against 0.661 at $\lambda = 1$. In the frontier arm Mistral Large cooperates more than Claude 3.5 Haiku at $\lambda = 0.1$ (0.502 against 0.468) and less at $\lambda = 1$ (0.538 against 0.700).

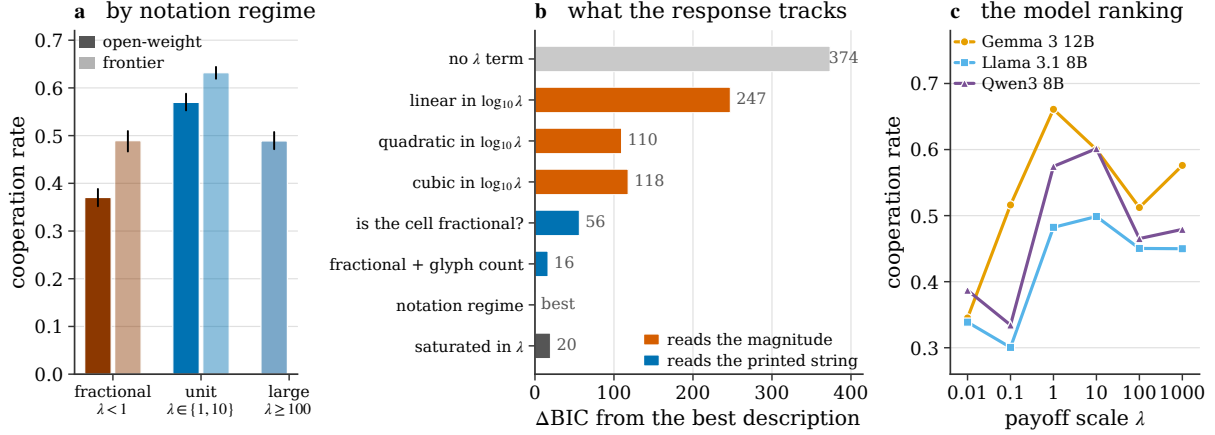


Figure 3: What the response is a function of. (a) Cooperation by notation regime, pooled within each arm. Whiskers are 95% dyad bootstrap intervals. The frontier arm has no large regime, because its sweep stops at $\lambda = 10$. (b) The description ladder for the open-weight arm. Bars give the BIC penalty each specification pays relative to the best one, with model, language and both personas controlled in every row. A three-level factor built from how the cells are printed beats every polynomial in $\log_{10} \lambda$ and beats the factor saturated in λ , which has three more parameters. (c) Cooperation by model across the sweep, on an evenly spaced scale index. Qwen3 8B cooperates more than Gemma 3 12B at $\lambda = 0.01$ and less at every larger scale, so which of the two is the more cooperative model is a statement about the units.

A benchmark that fixes one scale reports one side of these comparisons and has no way to signal that the other side exists.

3.4 The effect is set at the opening move

A cooperation rate is an average over a game, so a difference between two scales could enter anywhere in it. It enters at the first move (figure 4a,b). In the open-weight arm the gap between the fractional and unit regimes is -0.129 on round one alone, against -0.199 over the whole game, so the opening move already carries 65% of it. In the frontier arm the round-one gap is -0.199 against -0.142 over the whole game: there the effect is *larger* at the opening than on average, and shrinks as play proceeds, from -0.199 on round one to -0.095 over rounds seven to ten.

Round one is the round on which no history exists. The agent has seen the payoff table, its persona and the round index, and nothing else. Whatever the rescaling does, therefore, it does to the prior over actions that the prompt induces, and not to anything the agent learns from its opponent. The two arms then diverge in what happens afterwards, which we report as a difference rather than a finding: over thirty rounds the open-weight gap widens slightly, from -0.129 to -0.210 , while over ten rounds the frontier gap closes by half. Neither trajectory creates the effect, and the shorter horizon does not manufacture it.

3.5 What the rescaling does not move

A corpus in which every measure moves is a corpus one should be suspicious of. This one is not of that kind (figure 4c). Measured on the same games and against the same dyad-level null, the level of cooperation moves in both arms, and so do mutual cooperation, mutual defection,

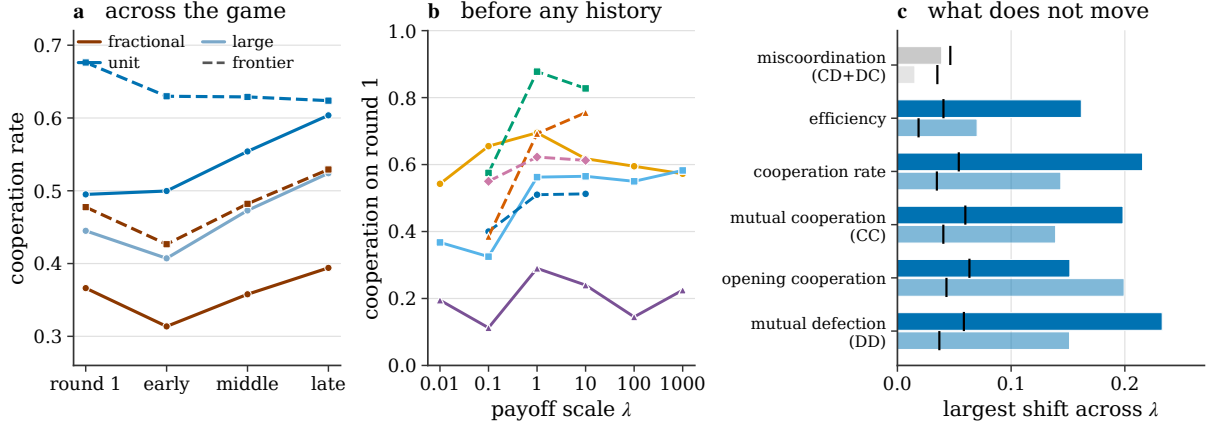


Figure 4: Where the effect enters, and what it leaves alone. (a) Cooperation by notation regime across the game. Solid lines are the open-weight arm, dashed the frontier arm; the four positions are round one and three equal blocks of the remaining rounds. The gap between the regimes is open at round one, before any history exists. (b) Opening-move cooperation against the payoff scale, one line per model, colours as in figure 2. (c) The largest shift across scales for six measures of the same games, against the same permutation null. Dark bars are the open-weight arm, pale the frontier arm; grey marks a measure whose shift does not clear its null. The level of cooperation moves in both arms; the rate at which the two agents fail to match does not.

efficiency and the opening move, all at $P < 0.001$. The rate of miscoordination, that is, the share of rounds in which the two agents chose differently, does not: it swings by 0.039 across six scales in the open-weight arm ($P = 0.14$, null 95th percentile 0.047) and by 0.015 across three in the frontier arm ($P = 0.56$, null 0.035), while mutual cooperation over the same rows swings by 0.198 and 0.139.

The distinction is worth stating plainly. Scaling the payoffs moves how much the agents cooperate and leaves untouched how well they coordinate. Two agents at $\lambda = 0.01$ and two at $\lambda = 1$ agree with each other equally often; what differs is what they agree on. The asymmetry between the two positions in a dyad behaves the same way in the open-weight arm, where the first-listed agent cooperates between 0.054 and 0.099 less than the second at every one of the six scales, with no trend across them.

3.6 The scale shifts how far the persona instruction is obeyed

Each agent received a system-prompt persona, either *cooperative* or *selfish*. A persona is not irrelevant by theorem, but it carries no information about the game: the four pairings present identical payoff matrices, and an agent was never told its opponent's persona. What the persona does carry is an instruction, and how far that instruction is obeyed turns out to depend on the payoff scale (figure 5a, table 1).

The result has to be stated at the level of the individual model, because the pooled average is misleading here in a way worth setting out. Pooled over the open-weight arm the effect of the cooperative persona is -0.095 $[-0.139, -0.051]$ at $\lambda = 0.01$, turns positive at $+0.113$ $[0.068, 0.159]$ at $\lambda = 10$ and collapses to -0.003 $[-0.049, 0.044]$ at 1000; the frontier pooled curve runs -0.461 $[-0.494, -0.429]$, $+0.030$ $[-0.005, 0.065]$ and $+0.073$ $[0.039, 0.109]$. Read alone, those two curves say that the instruction is inverted in the fractional regime, obeyed in the unit regime and inert in the large one. That reading does not survive disaggregation. In

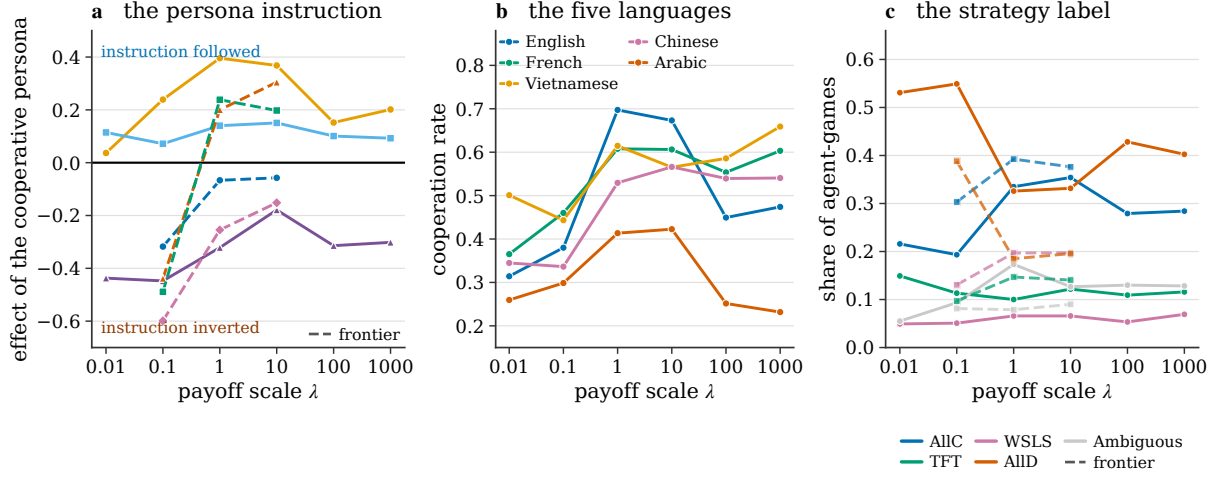


Figure 5: What the payoff scale carries with it. (a) The effect of being told to be cooperative rather than selfish, scale by scale, *one line per model*; colours as in figure 2, solid the open-weight arm and dashed the frontier arm. Every model moves upward across the fractional-to-unit boundary and the three open-weight models move back down into the large regime, but only two of the seven cross zero: the sign of the effect is a property of the model. The average of these lines crosses zero although almost none of them does, which is why no pooled curve is drawn. (b) Cooperation by prompt language across the sweep, open-weight arm. The five languages do not move together: Arabic falls from $\lambda = 10$ onward while Vietnamese rises. (c) The share of agent-games carrying each strategy label, against the payoff scale, with the arms encoded as in figure 4a. The two unconditional rules trace the cooperation curve of figure 2 and its mirror image; the two conditional rules stay nearly flat.

five of the seven models the effect keeps one sign across the entire sweep: it is positive at every scale for Gemma 3 12B and Llama 3.1 8B and negative at every scale for Qwen3 8B, Claude 3.5 Haiku and Mistral Large. Every one of those 21 estimates excludes zero except Gemma at $\lambda = 0.01$, and every one of the five is signed consistently once the scales are pooled within a regime (table 1). Only GPT-4o and Gemini 3.5 Flash-Lite cross zero. The pooled sign change is produced by three open-weight models whose constant-sign effects change in magnitude, not by any model reversing: Qwen3’s inversion weakens from -0.442 to -0.250 while Gemma’s compliance strengthens from $+0.138$ to $+0.382$. The apparent null in the large regime is the same arithmetic, since at $\lambda = 100$ the three open-weight effects are $+0.152$, $+0.101$ and -0.314 and each of the three excludes zero. Consistent with that, the three-way interaction between model, regime and persona is itself significant, at $F(3, \cdot) = 50.8$, $P = 2.7 \times 10^{-32}$ in the frontier arm and $F(4, \cdot) = 7.6$, $P = 4.2 \times 10^{-6}$ in the open-weight arm, so the pooled two-way interaction is not a summary of a pattern the models share.

What the models do share is the direction of the shift. Measured as a difference in differences with its own dyad bootstrap, the persona effect moves toward compliance across the fractional-to-unit boundary in every one of the seven, by $+0.244$, $+0.052$ and $+0.192$ in the open-weight arm and $+0.256$, $+0.691$, $+0.707$ and $+0.396$ in the frontier arm, and all seven intervals exclude zero. Where the sweep continues into the large regime all three open-weight models move back the other way, by -0.205 , -0.049 and -0.058 , the first two of which exclude zero. The persona effect therefore follows the same non-monotone course as the cooperation rate itself, rising into the unit regime and falling out of it, in seven of seven models and both arms.

The two statements have to be kept apart, and separating them is what the result is. *Whether* a model obeys a cooperative persona at all is a property of that model, stable across five orders

Table 1: The persona instruction, by model and notation regime. The effect is the cooperation rate of an agent told to be cooperative minus that of an agent told to be selfish, measured within a regime. “Shift” is the difference between two of those effects, with its own bootstrap over whole dyads. A dagger marks an entry whose 95% interval excludes zero. The last column records whether the effect keeps one sign over the whole sweep. Read down the shift columns: every model moves toward compliance entering the unit regime and away from it leaving the unit regime, which is the shared pattern. Read down the effect columns: five of the seven never change sign, which is why the pooled curve of §3.6 describes no individual model. The same effects scale by scale rather than regime by regime are in the electronic supplementary material, table S4.

Arm	Model	Effect of the cooperative persona			Shift		Sign
		fractional	unit	large	frac.→unit	unit→large	
open-weight	Gemma 3 12B	+0.138 [†]	+0.382 [†]	+0.177 [†]	+0.244 [†]	−0.205 [†]	constant
open-weight	Llama 3.1 8B	+0.093 [†]	+0.145 [†]	+0.097 [†]	+0.052 [†]	−0.049 [†]	constant
open-weight	Qwen3 8B	−0.442 [†]	−0.250 [†]	−0.308 [†]	+0.192 [†]	−0.058	constant
frontier	Claude 3.5 Haiku	−0.318 [†]	−0.062 [†]	-	+0.256 [†]	-	constant
frontier	Gemini 3.5 Flash-Lite	−0.489 [†]	+0.218 [†]	-	+0.707 [†]	-	crosses
frontier	GPT-4o	−0.438 [†]	+0.252 [†]	-	+0.691 [†]	-	crosses
frontier	Mistral Large	−0.599 [†]	−0.203 [†]	-	+0.396 [†]	-	constant

The frontier arm has no large regime, because its sweep stops at $\lambda = 10$. Model $\times$ regime $\times$ persona interaction: $F(3, \cdot) = 50.8$, $P = 2.7 \times 10^{-32}$ (frontier); $F(4, \cdot) = 7.6$, $P = 4.2 \times 10^{-6}$ (open-weight).

of magnitude in the payoff scale, and nothing in this corpus suggests the units decide it. *How strongly* it obeys is not a property of the model: it is shifted, in a common direction and by as much as 0.71, by a change to the payoff matrix that carries no instruction, no information about the task, and no game-theoretic content whatever. For two of the seven models that shift is large enough to carry the effect across zero, which is how the same system prompt comes to raise cooperation at one scale and lower it at another in the same model.

3.7 Language and objective framing

The prompt language interacts with the scale as well (figure 5b). In the open-weight arm the spread between the five languages ranges from 0.162 to 0.427 depending on which scale it is measured at, and the scale swing ranges from 0.191 to 0.383 depending on which language it is measured in (electronic supplementary material, table S8); the interaction between the notation regime and language is $F(8, \cdot) = 19.4$, $P = 3.5 \times 10^{-29}$. The frontier arm gives the same result at a smaller size, $F(4, \cdot) = 7.9$, $P = 2.3 \times 10^{-6}$. The languages do not move in parallel: cooperation in Arabic falls from 0.423 at $\lambda = 10$ to 0.232 at $\lambda = 1000$ while cooperation in Vietnamese rises from 0.565 to 0.659 over the same two scales.

One feature of the FAIRGAME templates makes a partial mechanism test available, and we use it rather than treat it only as a limitation. The templates do not state the objective identically in every language: English, French and Vietnamese tell the agent to minimise a penalty, while Chinese and Arabic tell it to maximise a reward, with the cell values still described as penalties. Language is therefore confounded with objective framing in this corpus, which means the five-language contrast is an omnibus test and we report it as one.

The confound is informative here because a natural account of the scale effect predicts something specific about it. If models were reading the displayed numbers as rewards to be

maximised rather than penalties to be minimised, and if raising λ made the largest attainable number easier to notice, then the direction of the scale effect should differ between the two framings. It does not. Under both framings the open-weight response has the same non-monotone shape, rising from 0.302 to 0.494 and falling to 0.386 under the maximise-framed languages, and rising from 0.393 to 0.640 and falling to 0.579 under the minimise-framed ones. Framing shifts the level of cooperation, by 0.136 in the open-weight arm and by only -0.007 in the frontier arm, and does not reverse or reshape the response to λ . The scale effect is therefore not a reward-reading effect, or at least not only one.

3.8 The strategy label moves, and it moves selectively

Cooperation rate is one summary. A second, and the one that feeds evolutionary models of mixed populations, is the strategy a read-out assigns to a whole run of play. We pass every agent-game through the hybrid read-out of §2.4, which proves a label from the four canonical memory-one rules where one of them exactly reproduces the run, records the run as ambiguous where several do, and otherwise takes the argmax of an LSTM trained on synthetic play from the same four. The mix over those five categories is the dependent variable (figure 5c).

It moves with the scale in all seven models. The largest total-variation distance between the mixes at two scales is 0.375 for Gemma, 0.388 for Llama and 0.278 for Qwen3, against dyad-level permutation nulls of 0.143, 0.115 and 0.123; and 0.263 for Claude, 0.338 for Gemini, 0.325 for GPT-4o and 0.125 for Mistral Large, against nulls of 0.115, 0.120, 0.103 and 0.098. Every one of the seven exceeds its null, at $P < 0.001$ for six and $P = 0.006$ for Mistral Large. Restricting the vocabulary to the four named rules, so that movement into and out of the ambiguous category cannot contribute, leaves all seven significant, at 0.205, 0.388, 0.275, 0.245, 0.304, 0.296 and 0.127 in the same order. One model changes materially under that restriction and we flag it: Gemma’s shift falls from 0.375 to 0.205, because almost half of its five-category distance is movement into the ambiguous bucket rather than between strategies. For the other six the two versions agree to within 0.04.

The shape of the movement is the substantive part, and a single distance cannot show it (figure 5c, table 2). Resolved into one curve per label, the open-weight AllD share runs 0.531, 0.549, 0.326, 0.332, 0.428 and 0.403 across the six scales and the AllC share runs 0.216, 0.193, 0.335, 0.354, 0.279 and 0.284. These are the cooperation curve of figure 2a and its mirror image: the same rise into the unit regime, the same plateau across a full order of magnitude, the same fall above $\lambda = 100$. The frontier arm reproduces the rising limb, AllD falling from 0.388 to 0.185 and AllC rising from 0.303 to 0.393 between $\lambda = 0.1$ and $\lambda = 1$. The notation result therefore holds on a read-out that consumes the whole sequence, and not only on an average over rounds.

The two conditional rules behave differently, and the contrast is the point. Over the open-weight sweep the TFT share moves by 0.049 and the WSLS share by 0.020, against the 0.223 of AllD and the 0.161 of AllC; measured against the same dyad-level null, TFT clears it narrowly ($P = 0.0015$, null 0.038) and WSLS does not clear it at all ($P = 0.36$, null 0.029). The frontier arm gives the same ordering, with TFT at 0.050 and WSLS at 0.067 against AllD’s 0.203. What the payoff scale moves is how often an agent is assigned an *unconditional* rule, and which one; how often it is assigned a rule that responds to its opponent barely moves. That is the invariant of §3.5 seen at the level of the sequence rather than the round: reciprocal play is what produces coordination, and reciprocal play is what the rescaling leaves alone.

Table 2: Each strategy label against the payoff scale. Shares are of all agent-games in the arm at that scale. “Swing” is the largest difference between two scales, tested against a permutation null that shuffles the scale label across whole dyads, exactly as in table 3. The predicted swing is zero on every row. The two unconditional rules move by three to eleven times what the two conditional ones do, and win-stay lose-shift over six scales is the one label whose movement does not clear its null. “Ambiguous” is not a strategy: it records that several rules reproduce the run of play equally well, so its row measures how nameable the play is rather than what was played.

Label	share of agent-games at $\lambda =$						across the sweep		
	0.01	0.1	1	10	100	1000	swing	null	P
<i>open-weight arm, six scales, 30 rounds</i>									
AllD	0.531	0.549	0.326	0.332	0.428	0.403	0.223	0.063	< 0.001
AllC	0.216	0.193	0.335	0.354	0.279	0.284	0.161	0.054	< 0.001
TFT	0.149	0.113	0.100	0.122	0.109	0.116	0.049	0.038	0.0015
WSLS	0.049	0.051	0.066	0.066	0.053	0.069	0.020	0.029	0.36
Ambiguous	0.055	0.093	0.173	0.127	0.130	0.128	0.118	0.052	< 0.001
<i>frontier arm, three scales, 10 rounds</i>									
AllD	-	0.388	0.185	0.196	-	-	0.203	0.041	< 0.001
AllC	-	0.303	0.393	0.376	-	-	0.089	0.039	< 0.001
TFT	-	0.097	0.147	0.141	-	-	0.050	0.029	< 0.001
WSLS	-	0.131	0.197	0.198	-	-	0.067	0.033	< 0.001
Ambiguous	-	0.081	0.079	0.090	-	-	0.011	0.032	0.69

Provenance of the labels (§2.4), as a share of all agent-games in the arm: provable rule 22.1% open-weight and 25.5% frontier, ambiguous 11.8% and 8.3%, LSTM above the 0.90 posterior floor 49.6% and 40.5%, LSTM below it 16.5% and 25.7%. None of these four shares is constant across the sweep; the breakdown by scale is in the electronic supplementary material, table S7, and the mix resolved by model in table S6.

Two further observations belong here because they qualify what the read-out can be asked to support. First, Mistral Large is the model whose cooperation rate did not move, and its strategy mix does, at $P = 0.006$ over the five categories and $P = 0.004$ over the four named rules. A model can cooperate at the same rate under two scales and still be assigned a different distribution over strategies, because the rate is an average and the label is a claim about the sequence; invariance on one summary is not invariance on the other. Second, the read-out’s own coverage is not constant across the sweep. The share of open-weight agent-games carrying a provable rule falls from 0.284 at $\lambda = 0.01$ to 0.172 at $\lambda = 10$ and recovers to 0.207 at $\lambda = 1000$, while the ambiguous share rises from 0.055 to 0.173; in the frontier arm provable coverage falls from 0.360 to 0.190 across three scales and the share below the posterior floor rises from 0.219 to 0.290. Play is at its least exactly nameable at the scales where cooperation is highest, which is a property of the instrument as much as of the behaviour, and it is a further reason to read the named-vocabulary column of table 3 alongside the five-category one rather than instead of it.

4 Discussion

4.1 A cooperation rate is a statement about one choice of units

The central result is that the strategic behaviour of a language model is not invariant to the one manipulation whose null is a theorem. Six of seven models move, by up to 0.32 in cooperation rate and up to 0.39 in total variation over strategy labels, when every payoff is multiplied by a

Table 3: Scale sensitivity per model. “Swing” is the largest difference in cooperation rate between two payoff scales and “at” names the scale where cooperation peaks. “Monotone” records whether the response increases or decreases throughout the sweep. “Coop. shift” repeats the swing as a test against a permutation null that shuffles the scale label across whole dyads, and “mix shift” is the same test on the largest total-variation distance between the strategy mixes at two scales, reported over the five categories the read-out emits and again over the four named rules alone, on which movement into the ambiguous category cannot contribute. “Best description” is the specification of table 4 with the lowest BIC for that model. The predicted value of every shift column is exactly zero, because a positive rescaling of a von Neumann–Morgenstern utility leaves the game unchanged. Nothing in this table is comparable across the two arms, whose horizon, base matrix and decoding settings all differ (§2.6).

Arm	Model	Cooperation				Coop. shift		Mix shift		
		Coop.	swing	at	Mono.	obs.	null	5-cat.	named	null
open-weight	Gemma 3 12B	0.535	0.316	$\lambda=1$	no	0.316***	0.121	0.375***	0.205***	0.143
open-weight	Llama 3.1 8B	0.420	0.198	$\lambda=10$	no	0.198***	0.047	0.388***	0.388***	0.115
open-weight	Qwen3 8B	0.474	0.267	$\lambda=10$	no	0.267***	0.095	0.278***	0.275***	0.123
frontier	Claude 3.5 Haiku	0.625	0.240	$\lambda=10$	yes	0.240***	0.058	0.263***	0.245***	0.115
frontier	Gemini 3.5 Flash-Lite	0.740	0.080	$\lambda=1$	no	0.080**	0.062	0.338***	0.304***	0.120
frontier	GPT-4o	0.448	0.234	$\lambda=10$	yes	0.234***	0.062	0.325***	0.296***	0.103
frontier	Mistral Large	0.524	0.036	$\lambda=1$	no	0.036	0.065	0.125**	0.127**	0.098

*** $P < 0.001$, ** $P < 0.01$, blank $P \geq 0.05$, over 2,000 dyad-level permutations; the null column is the 95th percentile of the five-category null, and the named-vocabulary nulls, which run from 0.094 to 0.130, are given in the electronic supplementary material, table S5. Best description by BIC: notation regime for Llama, Qwen3, Claude, Gemini and GPT-4o; a cubic in $\log_{10} \lambda$ for Gemma, the one model the notation account does not fit best; no λ term for Mistral Large.

positive constant. This is not a framing effect in the sense the term usually carries, because no description changed. The words of the prompt, the ordering of the options, the identity of the dominant action and the structure of the dilemma are all identical between two conditions that differ by a factor of ten.

The consequence for measurement is that a published cooperation rate for a language model is conditional on a design choice that the theory says is not a choice, and that is therefore almost never reported. The size of that conditioning is not small. In the open-weight arm the smallest within-model swing across the sweep exceeds the median spread between the three models at a fixed scale, and eight of forty-five model-pair comparisons change sign somewhere in the sweep. Two frontier models change places between $\lambda = 0.1$ and $\lambda = 1$. A field that compares models on cooperation is comparing them at whichever scale each paper happened to write down.

4.2 Notation, and what the account can and cannot claim

The response is not a smooth function of the magnitude. A cubic in $\log_{10} \lambda$ fits worse than a single binary feature of the printed string, and a two-parameter description built from how the cells are written matches what a free parameter per scale achieves. Read with the shape of the curve, which rises, plateaus and falls, the picture is a step function over notation regimes rather than a dose-response in the payoffs.

We are explicit about the limit of that claim. In this design $\lambda < 1$ and “the cells print with a decimal point” are the same event, so the results are equally consistent with a notation account and with an account under which sub-unit magnitudes are simply treated as small stakes. The

Table 4: The description ladder. Every specification predicts the cooperation rate of an agent-game from the same controls for model, prompt language, own persona and opponent persona, and differs only in how the payoff scale enters. ΔBIC is measured from the best specification within each arm; k counts the free parameters the specification adds to the controls. The saturated row is the ceiling on what any function of λ can fit, so a row that matches it at lower k has found the shape of the response. In the frontier arm only two notation regimes are present, so the fractional flag and the regime factor are the same specification, and a cubic in $\log_{10} \lambda$ is collinear with a quadratic over three scales.

How λ enters	open-weight, 6 scales		frontier, 3 scales	
	k	ΔBIC	k	ΔBIC
not at all	0	373.5	0	192.9
linear in $\log_{10} \lambda$	1	247.2	1	53.7
quadratic in $\log_{10} \lambda$	2	109.6	2	8.4
cubic in $\log_{10} \lambda$	3	117.9	-	8.4
fractional flag	1	56.1	1	0.0
fractional flag + glyph count	2	16.5	2	8.4
notation regime, 3 levels	2	0.0	1	0.0
saturated in λ	5	19.6	2	8.4

evidence that favours notation is indirect and comes from three directions. First, the response is flat between $\lambda = 1$ and $\lambda = 10$, a full order of magnitude, in both arms, which a stake-size account has to explain away. Second, the response turns down again above $\lambda = 100$, so it is not monotone in stakes in either direction. Third, most of the effect is already there on the opening move, which is where a reading of the prompt lives and not where an evaluation of accumulated stakes could be.

The decisive experiment is one restatement away and we did not run it. Print the $\lambda = 1$ matrix as “0.0, 2.0, 8.0, 10.0” rather than “0, 2, 8, 10”, which holds the magnitude fixed and changes only the notation, and print the $\lambda = 0.1$ matrix in a unit where the cells are integers, which holds the notation fixed and changes only the magnitude. The two cells of that design separate the accounts completely. We regard the present results as establishing that the response is a step function over how the matrix is written, and as leaving open which feature of the writing does the work.

4.3 An instruction is only as good as the numbers next to it

The persona result is the one we would put first in front of anyone building agent systems, and it needs stating in two parts because the two carry different weight.

The part that is common to every model is a shift. The strength with which a model follows a cooperative persona moves toward compliance when the payoffs next to that instruction are written as short integers rather than decimals, in all seven models, by between 0.05 and 0.71, and moves back away from compliance when they are written as thousands, in all three models whose sweep goes that far. Nothing about the instruction changed, nothing about the task changed, and nothing a game theorist can compute from the matrix changed. Instruction-following is normally treated as a property of a model, measured and reported as such; what this says is that it is a property of a model and its numeric context jointly, and that the relevant part of the context can be a part carrying no instruction and no information about the task beyond its units. An evaluation of whether a model follows a behavioural instruction, run at one payoff

scale, does not transfer to another scale of the same task.

The part that is *not* common to every model is the sign, and we were careful in §3.6 not to claim otherwise, because the pooled average invites it. Two of the seven models obey a cooperative persona at one payoff scale and invert it at another; the other five keep whatever disposition they have across five orders of magnitude. Whether a system prompt is followed or reversed therefore remains a model-level fact worth measuring once. How much it is followed is not, and cannot be reported without the scale it was measured at. A practitioner should read the first as a property to select a model on, and the second as a reason to test the prompt at the units the deployment will actually use.

4.4 Consequences for models of hybrid populations

Work that models populations in which artificial and human agents interact [2, 13, 14], and the evolutionary-game machinery it draws on [24, 25], assigns agents strategies from a known family and computes the resulting dynamics. Three of our results bear on that programme directly. First, the assignment is not stable under a transformation the dynamics cannot see: the replicator equation is identical at λ and 10λ , and the strategy mix we would feed into it differs by up to 0.39 in total variation, or 0.39 over the four named rules alone. Two population models that a game theorist would call the same model therefore predict differently about the same agent.

Second, the instability sits in exactly the coordinate such models are most sensitive to. What moves is the split between AllC and AllD, by 0.16 and 0.22 over the open-weight sweep, and the seeded proportion of unconditional cooperators and defectors is what sets whether a replicator trajectory runs to fixation or to coexistence. Third, and cutting the other way, the reciprocal strategies are close to invariant: TFT moves by 0.05 and WSLS by 0.02, and WSLS does not clear its null at all. Since it is the reciprocal share that governs whether cooperation can be sustained against invasion at all [3, 20], the part of the mix that carries the qualitative behaviour of the dynamics is the part the rescaling leaves alone, and the part it moves is the initial condition. That is a narrower problem than the raw total-variation number suggests, and a more tractable one.

The practical implication is not that the programme is unworkable. It is that a population model built on measured LLM strategies inherits a free parameter that the theory does not contain, that the parameter enters through the initial mix rather than through the reciprocal core, and that it should be swept rather than fixed.

4.5 Relation to the prompt-sensitivity and framing literatures

That language model outputs move with features of the prompt that carry no task content is established: formatting choices, separators and option order all shift accuracy [26, 27], and personas shift behaviour in ways that outrun what the persona states [28, 29]. Our contribution is not that a spurious feature matters, but that the feature here is one the relevant theory has already declared to be spurious with a proof. In an accuracy benchmark, the claim that a separator should not matter is an intuition. In a game, the claim that units should not matter is the representation theorem, and that is what makes the size of the failure interpretable rather than merely surprising.

The relation to the human framing literature runs the other way [30, 31]. Human framing

effects are produced by re-describing the same outcomes, and they map to a compact set of psychological mechanisms: reference points, loss aversion, the salience of a gain or a loss. Nothing is re-described here. Within an arm the two conditions differ by a multiplicative constant, which is the one change that prospect theory and expected utility theory agree in treating as inert once the units of the value function are fixed. Our closest neighbour in the LLM literature is the finding of Lorè and Heydari [12] that contextual framing carries weight comparable to game structure; we document a sensitivity one step further out, to a change that is not a framing at all.

4.6 Limitations

Six limitations bound these conclusions.

The notation and magnitude accounts are not separated, for the reason given in §2.3, and the experiment that would separate them is stated above. This bounds the interpretation of §3.2 and not the result of §3.1, which needs no account of the mechanism to stand.

We test the multiplicative half of the affine invariance and not the additive half. Adding a constant to every cell is the weaker of the two demands, since it does not change the number of digits, and we would expect a smaller effect; but it has a cleaner interpretation, because it moves the reference point without touching the notation, and it is the natural companion experiment.

The two arms differ in horizon, in horizon disclosure, in decoding temperature and in the base matrix, so nothing is comparable across them and we compare nothing. The replication we claim is of the direction and the shape of the response, both of which are measured within an arm and within a model.

The sweep is coarse. Six scales at decade spacing locate the maximum only to within a decade, and the three-level regime partition is a coarse summary of a response that a denser sweep might resolve differently. The claim that a smooth function of $\log_{10} \lambda$ is the wrong shape does not depend on the spacing; the claim that three regimes is the right partition does.

The strategy result rests on an instrument, and the instrument is imperfect in a way that is worth naming rather than absorbing. Two thirds of the labels are supplied by a learned classifier rather than by a proof, and a sixth to a quarter of all agent-games sit below the posterior floor at which a read-out designed to abstain would decline to answer (§2.4). We have reported every test on the four named rules as well as on all five categories, and the conclusions of §3.8 hold on both; but the exact-rule coverage of the frontier arm is lower than the classifier’s training distribution, so the frontier labels in particular are nearest neighbours more often than they are recognitions. What would remove the caveat is an abstaining read-out, reported with its coverage, rather than a larger vocabulary.

Finally, the model set is seven, none of them with an explicit reasoning procedure, and reasoning traces were disabled by design so that the parser could not manufacture cooperation. Whether a model that reasons at inference time recovers the invariance is the single most valuable extension, and it is the one experiment whose result we would not want to predict.

4.7 A reporting standard, and what it costs

The remedy is cheap, and it follows from the first result alone. Any measurement of strategic behaviour in a language model should report the scale at which the payoff matrix was written, and should report the measurement at more than one scale. One decade of separation is enough

to detect the effect (electronic supplementary material, table S9). Taking the single contrast $\lambda = 0.1$ against $\lambda = 1$, and holding it to the null of the full sweep, which is conservative because a two-level design generates a smaller one, the gap is 0.232, 0.226, 0.080 and 0.036 for the four frontier models and 0.145, 0.182 and 0.240 for the three open-weight ones, against nulls of 0.058, 0.062, 0.062, 0.065, 0.121, 0.047 and 0.095. Six of the seven clear it, and the one that does not is the model that does not move over the whole sweep either. Three conditions spanning two decades additionally recover the shape. The cost is a factor of two or three in sampling on a quantity that is otherwise reported as though it were a constant of the model.

Concretely, we propose that a cooperation rate be published alongside its *scale swing*, the largest difference between two scales, and its permutation null. On this corpus those pairs are 0.316 against 0.121, 0.198 against 0.047, 0.267 against 0.095, 0.240 against 0.058, 0.080 against 0.062, 0.234 against 0.062 and 0.036 against 0.065. Six of the seven exceed the null, and the seventh is a genuine and interesting negative that the current practice of reporting one number cannot express.

4.8 Conclusion

Seven language models played 6,000 iterated prisoner’s dilemmas in which the only manipulation of the game was to multiply every payoff by a positive constant, a transformation whose effect on any theory of the game is exactly zero. Cooperation moved in six of the seven, by up to 0.32, and the strategy mix moved in all seven. The response is not monotone in the payoff magnitude, and it is better described by how the numbers are printed than by any polynomial in their logarithm. It is already open on the opening move, so it is a prior the notation induces and not something learned from play. It shifts how far a persona instruction is obeyed, in the same direction in every model, and it interacts with the prompt language. What it leaves alone is reciprocity: how well the two agents coordinate, and how often either is playing a rule that responds to the other.

The reading we draw is not that these models fail to be rational agents, which would be a small claim about a well-known fact. It is that the quantity a benchmark reports is jointly determined by the game and by an arbitrary choice of units that the theory promised was free. Until that choice is swept rather than fixed, a cooperation rate attributed to a model is attributed too widely.

Ethics

This work involved no human participants and no animal subjects, and required no ethical approval. The agents studied are commercial and open-weight language models, queried through their providers’ interfaces under the respective terms of service. No personal data were collected, generated or processed at any stage: every prompt is a payoff matrix, a persona sentence and a history of prior moves, and every response is a single option token.

Data accessibility

The full corpus of 6,000 dyads and 264,000 logged decisions, the analysis pipeline that produces every figure and table in this paper and in the electronic supplementary material, and the

prompt templates are available at [REPOSITORY URL]. Every number appearing in a figure or a table is also written to a machine-readable CSV file in the same repository, and each figure and supplementary table is generated from those files rather than transcribed, so no displayed value can drift from the analysis that produced it. A permanent archive with a DOI will be deposited on acceptance.

Additional material is deposited with the manuscript as electronic supplementary material: §S1 there validates the strategy read-out, and tables S1–S10 give the per-model and per-cell breakdowns behind the summary statistics reported here.

Declaration of AI use

Large language models are the object of study of this paper rather than a tool used to produce it: the corpus consists of their logged behaviour, and the models, versions and decoding settings are stated in §2.2. [AUTHORS TO COMPLETE: declare here any use of generative AI in preparing the manuscript itself, for example in drafting, editing or code generation, in accordance with the journal’s policy.]

Authors’ contributions

[A.O.]: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing - original draft; [A.T.]: conceptualization, methodology, validation, supervision, funding acquisition, writing - review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed herein.

Conflict of interest declaration

The authors declare no competing interests.

Funding

[FUNDING STATEMENT]

Acknowledgements

We thank the FAIRGAME developers at the Luxembourg Institute of Science and Technology for releasing the multilingual game templates on which the language manipulation is built.

References

- [1] Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi: 10.1145/3586183.3606763.

- [2] Allan Dafoe, Yoram Bachrach, Gillian Hadfield, Eric Horvitz, Kate Larson, and Thore Graepel. Cooperative AI: machines must learn to find common ground. *Nature*, 593:33–36, 2021. doi: 10.1038/d41586-021-01170-0.
- [3] Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981. doi: 10.1126/science.7466396.
- [4] Robert Axelrod. *The Evolution of Cooperation*. Basic Books, New York, 1984.
- [5] Martin A. Nowak and Karl Sigmund. Tit for tat in heterogeneous populations. *Nature*, 355: 250–253, 1992. doi: 10.1038/355250a0.
- [6] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi: 10.1126/science.1133755.
- [7] William H. Press and Freeman J. Dyson. Iterated Prisoner’s Dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109(26):10409–10413, 2012. doi: 10.1073/pnas.1206569109.
- [8] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9: 1380–1390, 2025. doi: 10.1038/s41562-025-02172-y.
- [9] Steve Phelps and Yvan I. Russell. Investigating emergent goal-like behaviour in large language models using experimental economics. *arXiv preprint arXiv:2305.07970*, 2023.
- [10] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. Nicer than humans: How do large language models behave in the prisoner’s dilemma? In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, pages 522–535, 2025. doi: 10.1609/icwsm.v19i1.35829.
- [11] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9):e2313925121, 2024. doi: 10.1073/pnas.2313925121.
- [12] Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14:18490, 2024. doi: 10.1038/s41598-024-69032-z.
- [13] Ana Paiva, Fernando P. Santos, and Francisco C. Santos. Engineering pro-sociality with autonomous agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32, 2018. doi: 10.1609/aaai.v32i1.12215.
- [14] Fernando P. Santos, Francisco C. Santos, and Jorge M. Pacheco. Social norm complexity and past reputations in the evolution of cooperation. *Nature*, 555:242–245, 2018. doi: 10.1038/nature25763.
- [15] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, Nicholas R. Jennings, Ece Kamar, Isabel M. Kloumann, Hugo Larochelle, David Lazer, Richard McElreath, Alan Mislove, David C. Parkes, Alex Pentland, Margaret E.

- Roberts, Azim Shariff, Joshua B. Tenenbaum, and Michael Wellman. Machine behaviour. *Nature*, 568:477–486, 2019. doi: 10.1038/s41586-019-1138-y.
- [16] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 1944.
- [17] R. Duncan Luce and Howard Raiffa. *Games and Decisions: Introduction and Critical Survey*. Wiley, New York, 1957.
- [18] Author One and Author Two. Payoff scaling shapes cooperation in LLM agents across languages, 2026. Preprint by the present authors.
- [19] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. FAIRGAME: A framework for AI agents bias recognition using game theory. In *ECAI 2025 — Proceedings of the 28th European Conference on Artificial Intelligence*, pages 4097–4104. IOS Press, 2025. doi: 10.3233/FAIA251300.
- [20] Martin A. Nowak and Karl Sigmund. A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner’s Dilemma game. *Nature*, 364:56–58, 1993. doi: 10.1038/364056a0.
- [21] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.
- [22] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993.
- [23] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2): 461–464, 1978. doi: 10.1214/aos/1176344136.
- [24] György Szabó and Gábor Fáth. Evolutionary games on graphs. *Physics Reports*, 446(4–6): 97–216, 2007. doi: 10.1016/j.physrep.2007.04.004.
- [25] Matjaž Perc, Jillian J. Jordan, David G. Rand, Zhen Wang, Stefano Boccaletti, and Attila Szolnoki. Statistical physics of human cooperation. *Physics Reports*, 687:1–51, 2017. doi: 10.1016/j.physrep.2017.05.004.
- [26] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models’ sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.
- [27] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8086–8098, 2022. doi: 10.18653/v1/2022.acl-long.556.
- [28] Ameet Deshpande, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, and Karthik Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1236–1270, 2023. doi: 10.18653/v1/2023.findings-emnlp.88.

- [29] Leonard Salewski, Stephan Alaniz, Isabel Rio-Torto, Eric Schulz, and Zeynep Akata. In-context impersonation reveals large language models' strengths and biases. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [30] Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981. doi: 10.1126/science.7455683.
- [31] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.